



College AI Assistant Chatbot

Your Smart Companion for
★ **Bright Future College of Engineering** ★

Hello!
How can I
help you
today?



AI & TECHNOLOGIES USED

Fine-Tuned Gemini 2.5 Flash | Google Vertex AI | Google Gen AI SDK | FastAPI | Python
HTML | CSS | JavaScript | Docker | Google Cloud Run



An AI-powered college information assistant designed to provide quick, accurate, and intelligent answers about **Bright Future College of Engineering**.



Built using a **Fine-Tuned Gemini 2.5 Flash** model through **Google Vertex AI** and deployed on **Google Cloud Run**.

FINE-TUNED MODEL USED



Fine-Tuned Gemini 2.5 Flash

The model is fine-tuned with college-specific information to deliver accurate, relevant, and context-aware responses.



**Gemini
2.5 Flash**

TECHNOLOGIES USED



Fine-Tuned
Gemini 2.5 Flash



Google
Vertex AI



Google Gen
AI SDK



FastAPI



Python



HTML



CSS



JavaScript



Docker



Google
Cloud Run

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WHAT CAN YOU ASK?



Courses & Programs

Explore our courses and specializations.



Admissions

Learn about the admission process, eligibility and requirements.



Placements

Get details about placement support and last year's statistics.



Hostel Facilities

Information about hostel rooms, amenities and rules.



Campus Facilities

Discover our labs, library, sports, events and more.



Transport & Support

Find transport options and student support services.

KEY FEATURES



AI-powered college information chatbot



Comprehensive college information



Course and program details



Admission-related information



Placement support information



Hostel information



Campus facilities information



Transport and student support



Quick question buttons



Interactive chatbot interface



Fine-Tuned Gemini 2.5 Flash model



Vertex AI integration



Deployed using Google Cloud Run



Responsive web interface



EXAMPLE QUESTIONS



What courses are available?



What is the admission process?



Does the college provide hostel facilities?



Is placement assistance available?



How many students were placed last year?



Is transport available?



SYSTEM ARCHITECTURE



USER
Asks a question in the chatbot



WEB INTERFACE
HTML / CSS / JS
(Chat UI)



FASTAPI
Backend API
Processes the request



VERTEX AI
Fine-Tuned
Gemini 2.5 Flash
(AI Model)



AI RESPONSE
Relevant answer
is returned to the user

HOW IT WORKS

1 User asks a question

The user enters a question through the chatbot interface.

2 Question sent to FastAPI

The frontend sends the question to the FastAPI backend using an API request.

3 FastAPI communicates with Vertex AI

The backend sends the question to the Fine-Tuned Gemini 2.5 Flash model using Google Gen AI SDK.

4 AI generates the response

The model understands the question and generates an accurate, context-aware answer.

5 Response delivered to user

The answer is sent back through FastAPI and displayed in the chatbot interface.



Instant Answers
Get quick and accurate responses anytime.



Reliable Information
Powered by a Fine-Tuned Gemini 2.5 Flash model.



Trusted & Secure
Built on Google Cloud with enterprise-grade security.



Accessible Anywhere
Use on desktop or mobile from anywhere.



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contact.sugumarai@gmail.com